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Inventor(s)	Chen; Kung-Cheng et al.

COLLAPSIBLE WORKBENCH WITH LOCKING DEVICE

Abstract

A collapsible workbench includes a parallelepiped casing including two rails disposed on two inner surfaces of two sides respectively; and a door; a table assembly including two slides having a first slide member slidably disposed on the rail and a second slide member slidably disposed on an inner surface of the first slide member; and a multipart table including a rectangular first sub-table disposed on the first slide members, and a rectangular second sub-table hingedly secured onto the first sub-table; and two locking devices including a front handle having a forward extension; a rear lock having a concave member; and a pivot pivotably secured to either side of a rear end of the first sub-table. The rear lock is heavier than the front handle. A pivotal movement of the extension causes a front end of the rail to be locked or unlocked by the concave member.

Inventors: **Chen; Kung-Cheng (Taichung City, TW), Huang; Lung-Chuan (Taichung City, TW)**

Applicant: **E-MAKE Co.,Ltd.** (Taichung City, TW)

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Background/Summary

FIELD OF THE INVENTION

[0001] The invention relates to workbenches and more particularly to a collapsible workbench having a pivotal locking device for fastening the workbench in a fully extended position in use.

BACKGROUND OF THE INVENTION

[0002] Conventional workbenches occupy a large space in order to achieve a stable support effect. However, this design has problems (e.g., inconvenient movement and space occupation), which make it difficult for the workbench to easily move during operation. To the worse, it is bulky. And in turn, it adversely affects effectiveness of the precious space, utilization and arrangement of work flow. As a result, work efficiency is greatly decreased.

[0003] Thus, the need for improvement still exists.

SUMMARY OF THE INVENTION

[0004] It is therefore one object of the invention to provide a collapsible workbench comprising a parallelepiped casing including two rails disposed on two inner surfaces of two sides respectively; and a door hingedly secured to a lower edge of a front opening; a table assembly including two slides having a first slide member lidably disposed on the rail and a second slide member slidably disposed on an inner surface of the first slide member; a limit member disposed on front ends of the second slide members; and a multipart table including a rectangular first sub-table disposed on the first slide members, and a rectangular second sub-table hingedly secured onto the first sub-table wherein in a fully extended position the second sub-table is disposed on the second slide members, and stopped by the limit member; and two locking devices including a front handle having a forward extension; a rear lock having a concave member; and a pivot pivotably secured to either side of a rear end of the first sub-table and disposed above the rail wherein a pivotal movement of the extension is configured to cause a front end of the rail to be locked by the concave member of the rear lock.

[0005] The above and other objects, features and advantages of the invention will become apparent from the following detailed description taken with the accompanying drawings

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Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. **1** is a perspective view of a collapsible workbench according to the invention;

[0007] FIG. **2** is a perspective view of the collapsible workbench with the door open;

[0008] FIG. **3** is a perspective view of the collapsible workbench with the multipart table extended out of the workbench by pulling;

[0009] FIG. **4** is a view similar to FIG. **3** with the slide further extended out of the workbench by pulling in an unlocked position;

[0010] FIG. **5** is a schematic longitudinal sectional view of FIG. **4**;

[0011] FIG. **6** is a detailed view of the area in a rectangle of FIG. **5**;

[0012] FIG. **7** is a perspective view of the collapsible workbench with the multipart table fully extended and secured in a locked position;

[0013] FIG. **8** is a schematic longitudinal sectional view of FIG. **7**;

[0014] FIG. **9** is a detailed view of the area in a rectangle of FIG. **8**;

[0015] FIG. **10** is a perspective view of the collapsible workbench mounted on a cabinet; and

[0016] FIG. **11** is a schematic longitudinal sectional view of FIG. **10** with a joining portion of the casing and the cabinet shown in details.

DETAILED DESCRIPTION OF THE INVENTION

[0017] Referring to FIGS. **1** to **11**, a collapsible workbench **100** in accordance with the invention is mounted on a cabinet **70** and comprises a casing **10**, a table assembly **20**, and two locking devices **30** as discussed in detail below.

[0018] The casing **10** includes a space S; two rails **12** disposed on two inner surfaces of two sides respectively, the rail **12** having a front end **121** and a rear end **122**; an opening **11**; and a door **13** hingedly secured to a lower edge of the opening **11**.
[0019] The table assembly **20** includes two slides **22** having a first slide member **221** slidably disposed on the rail **12** and a second slide member **222** slidably disposed on an inner surface of the first slide member **221**; a limit member **220** disposed on front ends of the second slide members **222**; and a multipart table **21** secured to the slides **22**. Thus, the table assembly **20** may extend out of the space S in a use position or retract into the space S in a storage position.
[0020] The multipart table **21** includes a first sub-table **211** disposed on the first slide members **221**, and a second sub-table **212** hingedly secured onto the first sub-table **211**. The second sub-table **212** is disposed on the second slide members **222** by pivoting. At this position, the second sub-table **212** is engaged with and stopped by the limit member **220**. Thus, the multipart table **21** is fully extended.
[0021] The locking device **30** includes a front handle **31** having a forward extension **311**; a rear lock **32**; and a pivot **33** pivotably secured to either side of a rear end of the first sub-table **211** and above the front end **121** of the rail **12**.
[0022] As shown in FIGS. **5** and **6** specifically, a user may press down the extension **311** to counterclockwise pivot the locking device **30** about the pivot **33** until the locking device **30** is disposed horizontally and the front end **121** of the rail **12** is not locked by a concave member **321** of the rear lock **32** in an unlocked position as indicated by P**2**. Next, the user may push the table assembly **20** into the space S for storage.
[0023] As shown in FIGS. **8** and **9** specifically, after the table assembly **20** is fully extended, the rear lock **32** automatically clockwise pivots about the pivot **33** because the rear lock **32** is heavier than the front handle **31** until the concave member **321** of the rear lock **32** is urged by the front end **121** of the rail **12** in a locked position as indicated by P**1**. As a result, the table assembly **20** is locked.
[0024] A fastening device **50** includes a first member **51** formed with the cabinet **70** and a second member **52** formed with the casing **10**. The fastening device **50** may secure the casing **10** to the cabinet **70**.
[0025] While the invention has been described in terms of preferred embodiments, those skilled in the art will recognize that the invention can be practiced with modifications within the spirit and scope of the appended claims.

Claims

1. A collapsible workbench, comprising: a parallelepiped casing including two rails disposed on two inner surfaces of two sides respectively; and a door; a table assembly including two slides having a first slide member slidably disposed on the rail and a second slide member slidably disposed on an inner surface of the first slide member; a limit member disposed on front ends of the second slide members; and a multipart table including a rectangular first sub-table disposed on the first slide members, and a rectangular second sub-table hingedly secured onto the first sub-table; and two locking devices including a front handle having a forward extension; a rear lock having a concave member; and a pivot pivotably secured to either side of a rear end of the first sub-table and disposed above the rail.
2. The workbench of claim 1, wherein in a fully extended position the second sub-table is disposed on the second slide members and stopped by the limit member.
3. The workbench of claim 1, wherein the rear lock is heavier than the front handle.
4. The workbench of claim 1, wherein a clockwise pivotal movement of the forward extension is configured to cause a front end of the rail to be locked by the concave member.

5. The workbench of claim 4, wherein a counterclockwise pivotal movement of the forward extension is configured to unlock the front end of the rail.